

PCASTNet: A Physics-Constrained Adaptive Style Transfer Network for Sample Generation in Cross-Machine Small-Sample Fault Diagnosis

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Abstract—To compensate for the overlooked data realism and physical consistency in traditional cross-machine sample generation methods, this paper proposes a Physics-Constrained Adaptive Style Transfer Network for Sample Generation in Cross-Machine Small-Sample Fault Diagnosis (PCASTNet) for generating diagnostic samples of the monitored machine under small-sample conditions. PCASTNet employs style transfer to decouple fault content from a reference machine and machine style from the monitored machine. Moreover, it further integrates physical priors by introducing a band energy preservation constraint during sample synthesis. Built upon multi-scale wavelet transform, the framework consists of a wavelet encoder, an Adaptive Style Normalization (AdaSN) module, a wavelet decoder, and a multi-objective loss that jointly constrains content fidelity, style consistency, and energy preservation. Experimental results on two cross-machine scenarios demonstrate that PCASTNet can generate a large number of samples from limited monitored machine data and significantly improves diagnostic accuracy under small-sample conditions.

Index Terms—Small-sample fault diagnosis, cross-machine, sample generation, style transfer, signal processing

I. INTRODUCTION

Fault diagnosis has been extensively studied over the past decades, with a wide range of techniques developed across signal processing [1], machine learning [2], [3] in various fields such as power generation [4]–[7], aerospace [8], automated manufacturing [9], and transportation [10], [11]. Despite this progress, practical deployment in real-world industrial scenarios remains challenging, particularly in cross-machine

small-sample fault diagnosis settings, where the compounded effects of data scarcity and domain discrepancy significantly hinder diagnostic performance.

In many real-world settings, some machines operate under highly reliable conditions, resulting in only a few available fault samples [12]. Meanwhile, abundant historical data may exist on other similar machines operating under different structural tolerances [13], sensor layouts [14], and operational loads [15], [16]. These constraints give rise to the increasingly important research topic of cross-machine small-sample fault diagnosis, which aims to achieve accurate fault diagnosis on data-scarce machines by leveraging auxiliary knowledge from data-rich devices.

To tackle the problem of cross-machine fault diagnosis under small-sample conditions, a few studies have explored preliminary solutions from the model design perspective [17]–[19]. These approaches aim to improve diagnostic performance by enhancing the ability of models to generalize across machines under limited data conditions. However, the fundamental issue of insufficient fault samples remains unresolved. This limitation becomes particularly pronounced in scenarios involving highly diverse equipment and extremely scarce fault data, where model-centric methods alone are insufficient to overcome the data bottleneck.

Therefore, recent studies have increasingly investigated sample generation techniques to alleviate the problem of limited fault samples by producing realistic synthetic data tailored to this scenario [20], [21]. These synthetic samples serve as auxiliary data sources that enhance cross-machine fault diagnosis models. Existing research can be broadly divided into two main paradigms: physics-based synthesis and data-driven generation.

Physics-based methods attempt to reproduce fault signals by constructing analytical models of mechanical systems. Techniques such as Finite Element Analysis (FEA) [22] and Multi-Body Dynamics (MBD) simulations [23] are often used to model the dynamic responses of defective components under known operating conditions [24]. However, these methods suffer from critical limitations in practice. Their accuracy is highly dependent on precise structural parameters and boundary conditions, which are often unavailable or difficult to measure in real-world industrial settings. Furthermore, the inherent simplifications in any physical model inevitably create a “reality gap” between the synthesized signals and those captured from an actual machine. Given these practical

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hurdles, which limit their scalability and fidelity, the focus of recent research has progressively shifted towards more flexible data-driven approaches.

Data-driven methods, in contrast, exploit the powerful representation capabilities of deep generative models to learn the underlying data distribution directly from real samples [25]. This paradigm has been dominated by Generative Adversarial Networks (GANs) [26] and their variants such as Dual Learning GAN (DualGAN) [27], Cycle-Consistent GAN (CycleGAN) [28], Discovery GAN (DiscoGAN) [29], and No-Independent-Component-for-Encoding GAN (NiceGAN) [30]. These models have been successfully adapted for fault diagnosis; for example, CycleGAN has been used to generate time-frequency representations [31], NiceGAN to bridge simulation-experiment gaps [32], and DualGAN to enhance early-stage fault signals [33]. Despite their success, these methods still face two fundamental challenges that motivate our work. **First, ensuring data realism remains difficult;** existing models often struggle with mode collapse, generating samples that lack diversity or fail to capture the specific statistical style of the target machine. **Second, they lack physical consistency;** operating as black boxes without explicit physical priors, they cannot guarantee that the generated samples adhere to the fundamental energy distribution principles of real-world signals, which can undermine the reliability of any downstream diagnostic task.

To improve the realism of synthetic data and physical plausibility, we propose a Physics-Constrained Adaptive Style Transfer Network (PCASTNet) for generating fault samples on the monitored machine, which typically suffers from a scarcity of available fault data. Compared with physics-based methods that rely on theoretical models, and traditional GANs that generate samples from random noise, the proposed framework enables data augmentation of small-sample data from the monitored machine by adhering to a style-content separation principle.

To provide an intuitive understanding of this concept, we offer a detailed explanation supported by a visual illustration. We draw an analogy from the physical composition of vibration signals, separating fault-specific information ('content') from machine-specific signatures ('style'). To provide a concrete illustration, Fig. 1 visualizes this concept using Continuous Wavelet Transform (CWT) [34] spectrograms from the CWRU dataset [35], showing two severities of a rolling element (ball) fault under two different operating conditions (motor load 0 hp/speed 1797 rpm and motor load 3 hp/speed 1730 rpm).

As the figure demonstrates, we define the 'content' as the universal, fault-discriminative semantic feature. The spectrograms for the 0.007 inch ball fault (left column) consistently exhibit a characteristic "double impulse stripe" pattern, while those for the 0.021 inch ball fault (right column) consistently show a "single impulse stripe" pattern. Crucially, this local fault feature persists even when the operating condition changes.

Returning to our method, to implement this principle in practice, we design an Adaptive Style Normalization (AdaSN) module. This module adaptively fuses Instance Normalization (IN) [36] and Layer Normalization (LN) [37] using a learnable

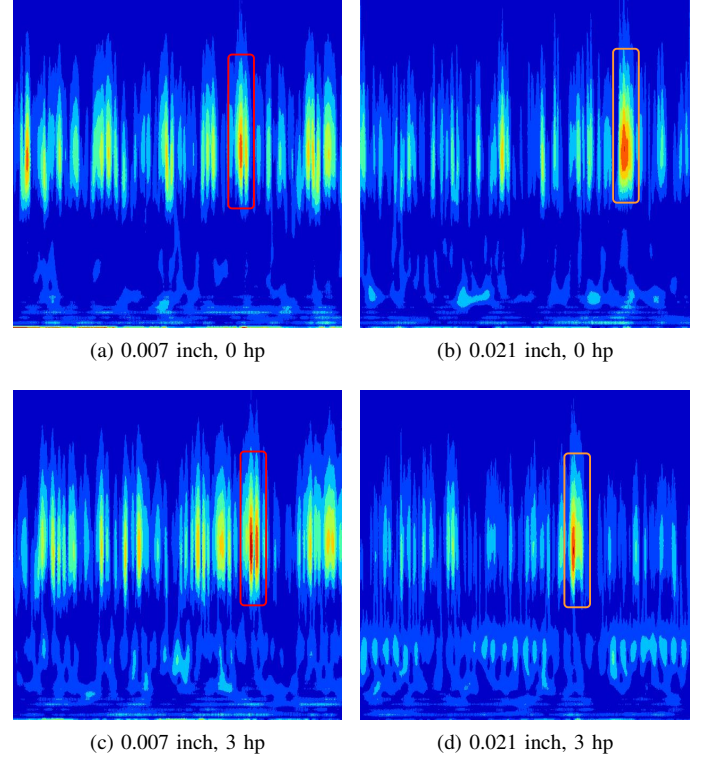


Fig. 1. Visual illustration of 'style' and 'content'. The semantic features within the red boxes (double impulse stripes) define the 'content' for the 0.007 inch ball fault, while those in the orange boxes (single impulse stripes) define the 'content' for the 0.021 inch ball fault. The variations between the top and bottom rows (e.g., in amplitude and texture) represent the 'style' change due to different operating conditions.

interpolation parameter ρ_s . By dynamically balancing the local texture preservation of IN and the global distribution alignment of LN through ρ_s , the module precisely extracts multi-scale style features of the monitored machine. Simultaneously, a wavelet encoder independently extracts content semantics via spectral encoding. This collaborative mechanism enables the generated samples to strictly follow the distribution characteristics of the monitored machine while faithfully preserving the physical semantics of the original fault. Furthermore, the integration of physics-aware constraints reinforces the spectral authenticity of the generated data during synthesis. As a result, the synthetic fault samples are inherently aligned with the vibration characteristics of the monitored machine, effectively expanding its training dataset and addressing the data scarcity challenge in cross-machine small-sample fault diagnosis.

The framework is built upon multi-scale wavelet transform, enabling the implicit injection of physical consistency without explicit differential equation modeling by leveraging AdaSN and a band energy preservation loss, thereby ensuring that the generated samples closely match real data.

The main contributions of this work are summarized as follows:

- 1) The proposed framework explicitly incorporates data realism and physical consistency—two aspects often overlooked by traditional sample generation methods. A style-content separation strategy preserves diagnostic

semantics and aligns with the monitored machine's statistical style, while a band energy preservation constraint maintains consistency with real signal energy distributions. These designs together improve diagnostic performance under cross-machine small-sample conditions.

- 2) We design an AdaSN module that combines the advantages of IN and LN via a learnable channel-wise coefficient ρ_s . This dynamic normalization strategy mitigates inter-machine style discrepancies and improves the flexibility of style alignment across different machines.
- 3) Extensive experiments conducted on multiple cross-machine small-sample tasks confirm that PCASTNet achieves superior diagnostic performance compared to existing baselines.

The remainder of this paper is organized as follows. Section II describes our methodology. Section III reports experimental results and analyses. Finally, Section IV concludes the work and outlines future directions.

II. METHODOLOGY

A. Framework Overview

To effectively characterize non-stationary fault signals, raw vibration inputs are first converted into time-frequency representations via CWT. This preprocessing step enhances transient feature extraction while mitigating spectral leakage issues inherent in Fourier-based methods [38], [39]. These representations then serve as input to the proposed PCASTNet, a novel framework for the generation of diagnostic samples between machines.

As illustrated in Fig. 2, PCASTNet consists of four tightly coupled modules:

- **Spectral Feature Encoder:** The encoder extracts multi-scale spectral features across frequency bands from the input time-frequency representations, effectively capturing both transient and stationary fault components.
- **AdaSN Module:** The encoder outputs are normalized by fusing IN and LN in a learnable manner. Specifically, a channel-wise parameter $\rho_s \in [0, 1]$ dynamically weights the IN and LN statistics to form adaptive mean-variance pairs. This normalization ensures that the fused features align with the monitored machine's statistical style while preserving fault content from the reference machine.
- **Feature Reconstruction Decoder:** The normalized features are decoded to reconstruct high-fidelity spectrograms. The decoder ensures both global structure and local spectral detail are preserved, enabling accurate signal interpretation for downstream diagnosis.
- **Multi-Loss Optimizer:** A multi-objective loss is applied to guide training. It consists of three terms: content fidelity to retain key diagnostic features, style consistency to match the monitored machine's distribution, and band energy preservation to inject physical realism. The last term enforces consistency in energy distribution across frequency bands, implicitly embedding mechanical prior knowledge.

The design rationale of this architecture is centered on a clear division of responsibilities to ensure effective style-

content decoupling. The Spectral Feature Encoder, pretrained on a large reference dataset and then frozen, acts as a universal extractor for diagnostic patterns ('content'). Its purpose is to recognize the fundamental spectral structures of faults, independent of the machine of origin. The core of the style transfer mechanism lies within the AdaSN Module. It operates by manipulating the channel-wise statistics (mean and variance) of the feature maps, which are known to encode textural and stylistic information. AdaSN first strips the original style from the content features via normalization and then imposes the new style by re-scaling the features using statistics derived from the monitored machine's samples. This process explicitly isolates the style manipulation from the content representation, allowing the fault pattern to be preserved while its appearance is altered.

Collectively, these modules establish an integrated workflow. The spectral feature encoder first extracts multi-scale fault characteristics from input time-frequency representations. Next, the AdaSN module dynamically aligns these features with the monitored machine's statistical signature, while preserving diagnostic semantics through adaptive fusion of IN and LN. The feature reconstruction decoder then synthesizes physically consistent spectrograms that maintain both global structural integrity and local spectral details. Concurrently, the multi-loss optimizer enforces content fidelity, style consistency, and band energy preservation through joint constraints. This cohesive workflow enables PCASTNet to generate machine-specific fault samples that retain physical plausibility and diagnostic relevance under limited data regimes, effectively augmenting scarce training data to significantly enhance cross-machine diagnostic performance without requiring architectural modifications for new machines.

Next, we detail these four modules in Sections II-B to II-E.

B. Spectral Feature Encoder

The spectral feature encoder is designed to extract fault-relevant spectral features from the input signals. Specifically, a VGG19-based classifier [40] is first pretrained on the reference dataset to learn general spectral patterns. During this pretraining stage, the network captures hierarchical features where shallow layers detect fine-grained edges and transients, while deeper layers encode higher-level spectral structures. Layers before ReLU4_1 are then extracted and used as the encoder. These layers are frozen during subsequent training to provide a stable feature representation. The resulting compact latent features are forwarded to the AdaSN module for adaptive style normalization, enabling style transfer while preserving fault-relevant spectral information.

C. AdaSN Module

Traditional normalization strategies, such as IN and LN, often exhibit limited flexibility in modulating feature statistics.

To better understand their roles in style transfer for fault diagnosis, we first analyze their individual behaviors. IN computes statistics independently for each channel, enabling fine-grained control over local texture, but tends to erase spatial structure that is critical for identifying localized fault patterns.

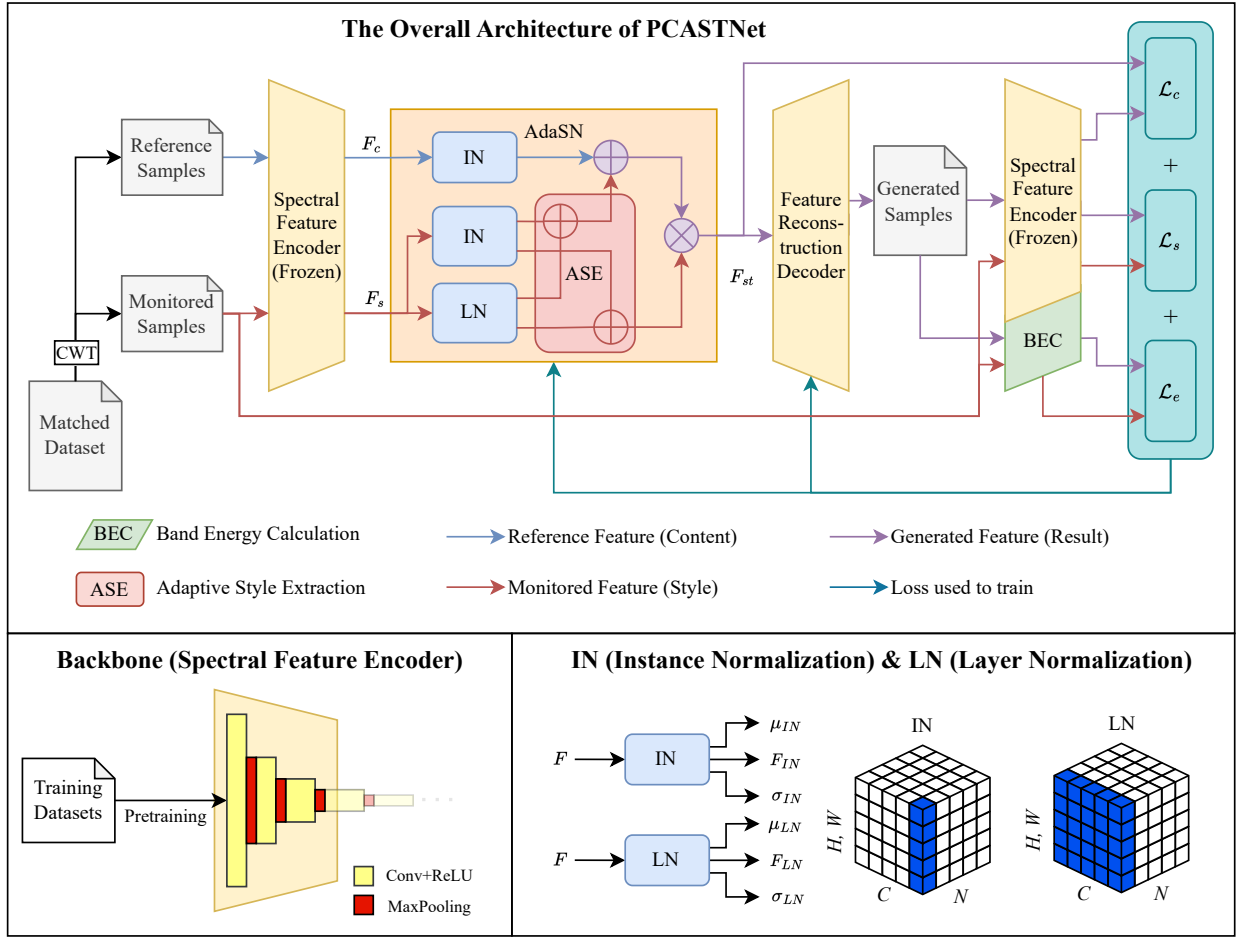


Fig. 2. The overall architecture of PCASTNet. Raw vibration signals from the reference and monitored machines are first transformed into time–frequency spectrograms via the CWT. A shared wavelet encoder extracts multi-scale spectral features. AdaSN aligns the fault content from the reference with the machine style of the monitored machine. The decoder reconstructs high-fidelity synthetic samples, guided by a multi-objective loss that enforces content fidelity, style consistency, and physical realism through band energy preservation.

In contrast, LN aggregates statistics globally across all channels, promoting stylization consistency but possibly weakening discriminative semantic cues. Mathematically, given a batch of input style features $F_s \in \mathbb{R}^{N \times C \times H \times W}$, where N is the batch size, and C , H , and W are the channels, height, and width of the feature maps, respectively, the normalized output by IN is given by:

$$F_{IN} = \frac{F_s - \mu_{IN}(F_s)}{\sqrt{\sigma_{IN}^2(F_s) + \epsilon}}, \quad (1)$$

where μ_{IN} and σ_{IN}^2 denote the mean and variance over spatial dimensions (H, W) for each channel and ϵ is a small constant added for numerical stability.

In contrast, LN normalizes all feature values within the same layer jointly:

$$F_{LN} = \frac{F_s - \mu_{LN}(F_s)}{\sqrt{\sigma_{LN}^2(F_s) + \epsilon}}, \quad (2)$$

where μ_{LN} and σ_{LN}^2 are computed across all channels and spatial positions.

These limitations have motivated the development of adaptive normalization strategies. One representative approach is Adaptive Instance Normalization (AdaIN) [41], which aligns

the channel-wise mean and variance of content features to those of style features. It has shown effectiveness in image generation and style transfer tasks, and can be expressed as:

$$\text{AdaIN}(F_c, F_s) = \sigma(F_s) \cdot \left(\frac{F_c - \mu_{IN}(F_c)}{\sigma_{IN}(F_c)} \right) + \mu(F_s), \quad (3)$$

where F_c and F_s denote content and style features, respectively, and μ and σ compute channel-wise statistics.

However, AdaIN relies solely on instance-level statistics and assumes uncorrelated feature channels, which can lead to loss of structural semantics essential for fault pattern recognition. To address this issue, we propose the AdaSN module that combines both instance-wise and layer-wise statistics, modulated by a learnable interpolation parameter ρ_s . This design improves the flexibility of style modulation while preserving task-relevant semantic information. Specifically, given content features F_c and style features F_s , the AdaSN transformation is defined as:

$$\text{AdaSN}(F_c, F_s) = \gamma_s(F_s) \cdot \left(\frac{F_c - \mu_{IN}(F_c)}{\sigma_{IN}(F_c)} \right) + \beta_s(F_s). \quad (4)$$

The affine parameters $\gamma_s(F_s)$ and $\beta_s(F_s)$ are dynamically generated from the style features via the Spectral Feature

Encoder:

$$\gamma_s(F_s) = (1 - \rho_s)\sigma_{\text{IN}}(F_s) + \rho_s\sigma_{\text{LN}}(F_s). \quad (5)$$

$$\beta_s(F_s) = (1 - \rho_s)\mu_{\text{IN}}(F_s) + \rho_s\mu_{\text{LN}}(F_s). \quad (6)$$

To enable fine-grained style modulation in normalization, we introduce an Adaptive Style Extraction (ASE) module within AdaSN. ASE generates a learnable interpolation parameter ρ_s to control the relative contribution between IN and LN. This factor allows the model to adaptively emphasize local stylization or global structure preservation depending on the diagnostic context. To ensure ρ_s remains a valid interpolation coefficient, it is constrained to the interval $[0, 1]$ after each training update. This is achieved via a clipping operation:

$$\rho_s \leftarrow \text{clip}_{[0,1]}(\rho_s - \tau \cdot \Delta\rho_s), \quad (7)$$

where the $\text{clip}_{[0,1]}$ function is defined as $\max(0, \min(1, x))$. τ denotes the learning rate, and $\Delta\rho_s$ is the gradient update vector computed by the optimizer.

This adaptive formulation enables the network to dynamically select the most suitable normalization behavior at each layer. When structural fidelity is prioritized, the model learns to reduce ρ_s , placing more emphasis on LN. Conversely, when strong stylization is desired, higher ρ_s favors IN-like behavior. The flexibility of AdaSN enables seamless generalization to varying machine structures and speed profiles.

Furthermore, AdaSN supports style transfer with extremely limited data while preserving both diversity and fidelity of the generated samples. Given n style and m content samples, AdaSN allows generation of up to $n \times m$ unique spectrograms. This combinatorial expansion substantially enhances the training corpus without additional real-world data, making PCASTNet an essential component for scalable small-sample fault diagnosis.

D. Feature Reconstruction Decoder

The decoder reconstructs stylized feature representations back into image samples for training downstream fault diagnosis models. It follows a symmetric architecture with respect to the encoder and replaces all pooling layers with nearest-neighbor up-sampling to mitigate checkerboard artifacts. Reflection padding is applied consistently in both encoder and decoder to suppress boundary artifacts during convolution operations.

Regarding normalization layers, IN aligns each sample to a single target style, while Batch Normalization (BN) centers a batch of samples around the same style. Both strategies constrain the model's capacity to generate images with diverse styles, which is particularly detrimental in tasks where style diversity is essential.

Consequently, we omit normalization layers in the decoder to preserve stylistic variability and avoid unintended style homogenization. The output images effectively retain diagnostic-relevant structures while integrating the visual characteristics of the target style, thereby enhancing the interpretability of subsequent diagnosis tasks.

E. Multi-Loss Optimizer

To ensure that the generated images preserve diagnostic semantics, exhibit the desired style characteristics, and maintain physically meaningful signal properties, we formulate a composite loss function comprising three components: content loss, style loss, and band energy loss. Each component addresses a distinct requirement of the style transfer process for fault diagnosis.

a) *Content Loss*: The content loss encourages the decoder to preserve the semantic structure of the input. Instead of comparing the output with the original content features $f(c)$, we use the PCASTNet-aligned feature map t as the content target. This design choice aligns the optimization objective with the stylized latent representation and improves convergence:

$$\mathcal{L}_{\text{content}} = \|f(g(t)) - t\|_2^2, \quad (8)$$

where $f(\cdot)$ denotes the frozen spectral feature encoder and $g(\cdot)$ is the feature reconstruction decoder. By minimizing the deviation between the decoded image and the AdaSN-normalized target, this loss reinforces semantic fidelity while decoupling it from the raw input content.

b) *Style Loss*: The style loss promotes alignment between the style image and the generated image in terms of appearance and texture. Inspired by AdaIN, we adopt a strategy based on channel-wise instance statistics instead of gram matrices [42]. This choice provides better interpretability. Specifically, we match the first- and second-order statistics (mean and standard deviation) of feature maps extracted from multiple layers of a pretraining-VGG19:

$$\begin{aligned} \mathcal{L}_{\text{style}} = & \sum_{i=1}^L \|\mu(\phi_i(g(t))) - \mu(\phi_i(s))\|_2 + \\ & \sum_{i=1}^L \|\sigma(\phi_i(g(t))) - \sigma(\phi_i(s))\|_2, \end{aligned} \quad (9)$$

where $\phi_i(\cdot)$ indicates the i -th layer activation. We select layers from ReLU1_1 to ReLU4_1 to form a multi-scale representation of style. This hierarchical approach is crucial because shallow layers capture low-level features like textures and colors, while deeper layers encode more abstract, global stylistic elements. By matching statistics across this entire hierarchy, our style loss ensures a comprehensive style transfer. Unlike Gram-based style losses, which may introduce texture redundancy and over-stylization, this formulation preserves both global and local appearance features in a controllable way.

c) *Band Energy Loss*: To enforce physical consistency in the generated samples, we introduce a band energy loss. The diagnostic-relevant information of a bearing fault is primarily encoded in its characteristic defect frequencies and their harmonics. However, in cross-machine tasks, distributional shifts caused by differing bearing models and operating conditions (e.g., rotational speed) pose a significant challenge. For instance, in the CWRU to BJTU-RAO [43] task, the CWRU dataset uses SKF 6205 bearings at speeds around

1730-1797 rpm (approximate 29-30 Hz), while the BJTU-RAO dataset uses HRB 352213 bearings at speeds up to 20, 40, 60 Hz. Although a fault may be of the same type, the different geometries and speeds result in distinct characteristic frequency distributions. A simple style transfer might align the visual texture but would fail to correct the underlying energy distribution in the frequency domain, leading to physically implausible samples.

The band energy loss directly addresses this problem. It ensures that the energy distribution of a generated sample along the frequency axis matches that of the target machine's style samples. This is critical for aligning the characteristic fault frequencies after style transfer, thereby ensuring the generated data is not just visually similar but also physically congruent with the target domain.

To formalize this, we first define the band energy distribution, E_{freq} , for a given spectrogram. It is calculated by summing the spectrogram's magnitude values along the time axis for each frequency bin, effectively projecting the time-frequency representation onto the frequency domain. This is computed as:

$$E_{\text{freq}}(f) = \sum_t |T(f, t)|, \quad (10)$$

where $T(f, t)$ is the time-frequency representation (i.e., the spectrogram), with f denoting the frequency axis and t denoting the time axis. The resulting vector $E_{\text{freq}}(f)$ represents the total energy present at each frequency band across the entire time window.

The loss is then defined as the L1 distance between the band energy distributions of the generated and target style images:

$$\mathcal{L}_{\text{energy}} = \|E_{\text{freq}}^{\text{gen}} - E_{\text{freq}}^{\text{style}}\|_1, \quad (11)$$

where $E_{\text{freq}}^{\text{gen}}$ and $E_{\text{freq}}^{\text{style}}$ are the band energy distribution vectors computed from the generated spectrogram and the target style spectrogram, respectively.

By imposing this constraint, we preserve essential spectral components like harmonics and sidebands that are crucial for diagnosis, making the model particularly effective in scenarios with variations in speed and bearing structure, as is the case across our three datasets.

d) *Total Loss*: The total objective is defined as a weighted combination of the above three losses:

$$\mathcal{L}_{\text{total}} = \lambda_c \mathcal{L}_{\text{content}} + \lambda_s \mathcal{L}_{\text{style}} + \lambda_e \mathcal{L}_{\text{energy}}, \quad (12)$$

where λ_c , λ_s , and λ_e are hyperparameters that control the balance among content fidelity, style alignment, and physical consistency. These weights can be tuned based on dataset conditions and diagnostic emphasis, particularly to preserve frequency-domain integrity in small-sample and cross-machine settings.

III. EXPERIMENT STUDIES AND RESULTS ANALYSIS

A. Datasets

To support small-sample fault diagnosis sample generation, as shown in Fig. 3, three publicly available bearing datasets are utilized: CWRU [35], HUSTbearing [44], and BJTU-RAO [43]

datasets. These datasets include four target labels: Normal (N), Inner Race Fault (I), Outer Race Fault (O), and Ball Fault (B), providing rich conditions and sufficient diversity for training and evaluating models under small-sample settings.

1) *CWRU Dataset*: The CWRU dataset, provided by Case Western Reserve University, is one of the most extensively used public benchmarks in bearing fault diagnosis. The testbed consists of a motor-shaft-bearing-load assembly equipped with multi-channel accelerometers, supporting various combinations of load and rotational speed. This dataset is well-structured with clearly defined fault types, complete annotations, and consistent acquisition settings, making it highly suitable for small-sample fault recognition and style transfer validation tasks.

Vibration signals are acquired from an accelerometer mounted on the drive-end bearing (DE channel). In this work, we select a 12 kHz sampling-rate subset and include normal baseline measurements to form the dataset. The selected signals are recorded under 0 horsepower load at three approximate speeds (1797, 1772, and 1750 rpm), covering fault diameters of 0.007, 0.014, and 0.021 inches.

To preserve temporal structure, raw signals are segmented into fixed-length samples of 1024 points using a sliding window approach. Each segment is labeled as one of: N, I, O or B.

2) *BJTU-RAO Dataset*: The BJTU-RAO dataset, provided by Beijing Jiaotong University, is the first publicly available dataset for fault diagnosis of railway bogie transmission systems. The data are collected on a testbed that replicates real-world subway operating conditions, covering diverse system-level fault modes. The BJTU-RAO dataset features challenging conditions including speed variation, system-level coupled faults, and real component damage, making it a rigorous testbed for evaluating sample generation and fault transfer methods under complex scenarios.

For this machine scenario, data from the left axlebox under four representative health conditions—N, I, O and B—are selected to construct a four-class fault identification task. The vibration signals are collected via an accelerometer mounted on the axlebox end cap (channel CH17), with a sampling rate of 64 kHz. Three motor speeds under no lateral load—20 Hz, 40 Hz, and 60 Hz—are used for analysis.

Each signal segment consists of 1024 data points and is used for model training and evaluation after preprocessing. All data are collected from the left axlebox and labeled as N, I, O or B accordingly.

3) *HUSTbearing Dataset*: The HUSTbearing dataset was collected by Huazhong University of Science and Technology using the Spectra-Quest mechanical fault simulation platform, which is designed to replicate realistic bearing fault scenarios encountered in industrial applications. The experimental setup consists of a variable-speed motor, a drive shaft, ER-16K bearings, acceleration sensors, and a data acquisition system. This dataset is characterized by diverse rotational speeds, clearly defined labels, and representative fault types, making it suitable for studies on cross-speed fault recognition and style transfer in small-sample conditions.

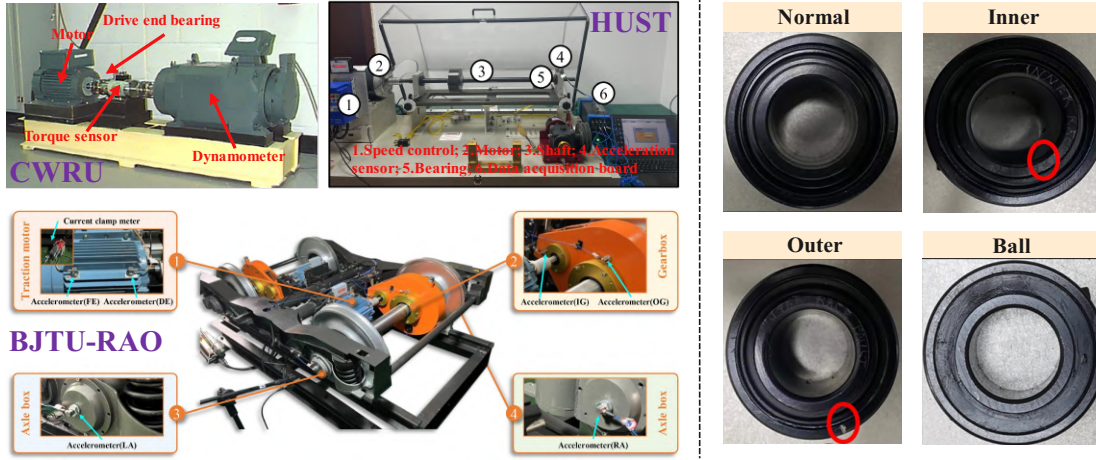


Fig. 3. Testing experimental platforms for different datasets.

The complete dataset includes nine health conditions and eleven working scenarios, covering both multiple fixed-speed and one variable-speed conditions. In this paper, we select five fixed-speed conditions—20 Hz, 40 Hz, 60 Hz, 70 Hz, and 80 Hz. All vibration signals were recorded via the X-direction acceleration channel at a sampling rate of 25.6 kHz. Each original signal contains 262,144 sampling points, corresponding to approximately 10.2 seconds of vibration data. Four fault types are considered under each selected speed: N, I, O and B. Each fault condition includes moderate and severe damage levels, with typical defect diameters ranging from 0.15 mm to 0.5 mm, depending on the specific location. To generate training samples, we apply a sliding window of length 1024 without overlap to segment the raw time series. The final sample labels are unified as N, I, O or B.

These datasets cover typical bearing fault modes including N, I, O and B under diverse operational conditions and machine backgrounds, providing comprehensive data support for small-sample generation and transfer learning in fault diagnosis.

B. Experimental Settings

The proposed small-sample generation framework consists of three main stages: data preparation, training process, and test process, as illustrated in Fig. 4.

a) *Data Preparation*: Raw time-domain vibration signals from both reference and monitored machines are first transformed into time–frequency spectrograms via wavelet transform. These spectrograms serve as the inputs for the encoder and the style transfer model.

The continuous wavelet transform of a signal $x(t)$ with respect to a mother wavelet $\psi(t)$ is defined as:

$$\mathcal{W}_x(a, b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} x(t) \psi^* \left(\frac{t-b}{a} \right) dt, \quad (13)$$

where a and b represent the scale and translation parameters, respectively, and $\psi^*(\cdot)$ is the complex conjugate of the mother wavelet. A larger scale a corresponds to lower-frequency components, while a smaller a captures high-frequency details.

In this work, we adopt the complex Morlet wavelet (‘cmor1-1’) due to its excellent time–frequency localization properties. The pseudo-frequency f_a corresponding to scale a is computed as:

$$f_a = \frac{f_c}{a}, \quad (14)$$

where f_c is the center frequency of the wavelet. This relationship enables physically interpretable mapping of wavelet coefficients to characteristic fault frequency bands.

As illustrated in Fig. 5, the raw vibration signals are segmented into fixed-length windows (e.g., 1024 points), standardized, and then converted into 2D spectrograms via CWT. The resulting amplitude scalograms are resized to a fixed resolution (e.g., 224×224) and color-mapped using the jet scheme to enhance frequency-band contrast. These spectrograms provide rich visual representations of time–frequency energy distributions. The high-scale components primarily capture low-frequency trends (e.g., imbalance, misalignment), whereas the low-scale components emphasize high-frequency impulses. This multi-scale representation enhances the network’s sensitivity to both slowly evolving and abrupt signal features, supporting comprehensive fault characterization.

For all experiments, 500 samples from the reference dataset (e.g., CWRU) are selected as content images, while 50 samples from the monitored dataset (e.g., BJTU-RAO or HUST-bearing) are used as style images. The content set conveys fault semantics, while the style set defines the machine-specific spectral characteristics. The choice of 50 style samples was deliberate, as these samples are not used for a single condition but are distributed across four major categories, which themselves include variations in bearings and rotational speeds. This results in an average of just over 12 samples per category, firmly establishing our experiment within a small-sample paradigm. This setting is consistent with, and in some cases even more challenging than the established definition of “small-sample” in recent state-of-the-art literature. Therefore, our experimental design represents a reasonable and well-justified small-sample setting. While our primary experiments use this multi-sample approach to ensure style extraction across diverse conditions, it is theoretically feasible

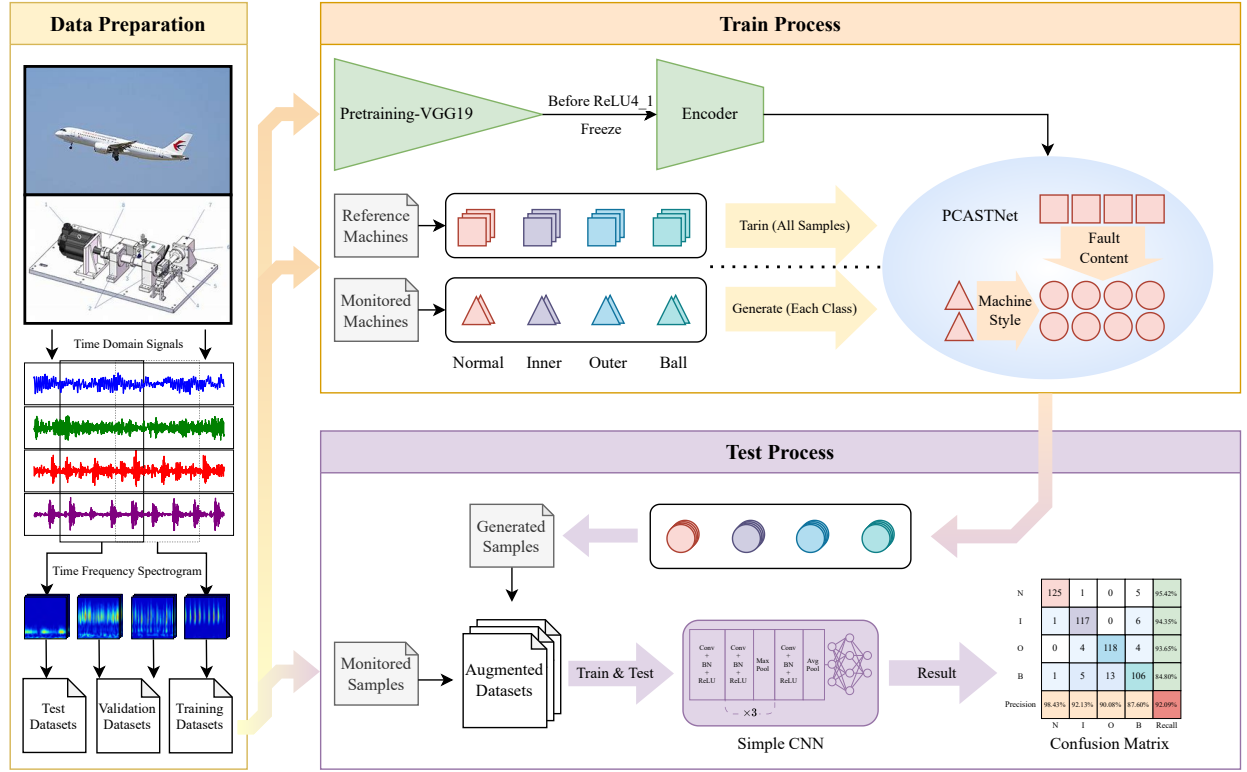


Fig. 4. The overall training and inference pipeline of the proposed PCASNet-based small-sample generation strategy. The framework consists of three stages: (i) **Data Preparation**, where raw vibration signals from reference and monitored machines are converted into wavelet-based time-frequency spectrograms; (ii) **Train Process**, including encoder pretraining on the reference machine and style transfer using PCASNet to generate style-aligned synthetic samples guided by monitored machine statistics; and (iii) **Test Process**, which serves as a rigorous protocol to evaluate the quality of our generated samples. This protocol evaluates the best-performing classifier, trained on the augmented dataset, against an unseen test set. The final performance is quantified using metrics such as overall accuracy and a detailed confusion matrix, providing a comprehensive assessment of the generated data's quality.

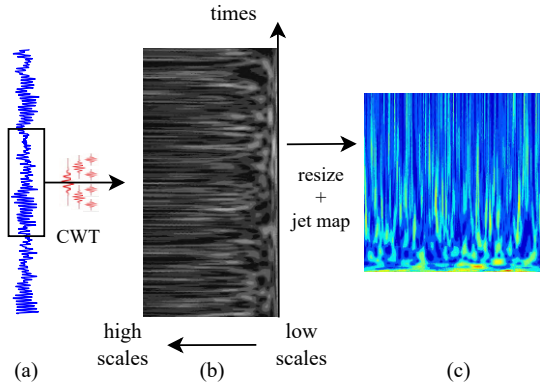


Fig. 5. The CWT-based signal preprocessing pipeline. (a) Raw vibration signals are first segmented into fixed-length windows (e.g., 1024 points) and standardized. (b) Each segment is transformed into a time-frequency spectrogram via CWT, using a complex Morlet wavelet and multi-scales. (c) The resulting amplitude scalograms are resized to a fixed resolution and color-mapped using the jet colormap to enhance scale-dependent contrast. The final spectrograms preserve both low-frequency trends and high-frequency transients, serving as interpretable inputs for model training.

for PCASNet to perform one-shot style transfer in scenarios with simpler, more consistent operating conditions.

b) Training Process: The training stage includes two key steps:

- **Encoder Pretraining:** A VGG19-based classifier is pre-trained on the reference dataset. Layers before ReLU4_1 are extracted as the encoder and frozen during subsequent training.
- **Style Transfer Training:** PCASNet is trained to generate spectrograms consistent with the monitored machine's style. Using the fixed encoder and the AdaSN module, 450 style-transferred samples are generated across all classes.

Tab. I summarizes the detailed layer composition of the proposed model for fault sample generation. The model-generated samples, in combination with the initial 50 monitored samples, yield an augmented training set comprising 500 instances.

c) Test Process: A simple CNN classifier is trained on the augmented dataset. To guide this training and prevent overfitting, we use 100 real small-sample instances from the monitored machine as a **validation set**, with another 500 instances reserved as the final **testing set**. The primary role of the validation set is to enable model selection through an early stopping strategy. Specifically, we monitor the classifier's performance on the validation set periodically during training (e.g., every 10 epochs). The training is terminated when the validation accuracy ceases to improve, and the model checkpoint that achieved the highest validation accuracy is saved as the best model. This best model is then evaluated on the unseen testing set. This entire stage evaluates the gener-

TABLE I
DETAILED ARCHITECTURE OF THE PROPOSED STYLE TRANSFER MODEL

Module	Layer	Operation
Encoder	Conv1	Conv(3, 64, 3×3) + ReLU
	Conv2	Conv(64, 64, 3×3) + ReLU
	Pool1	MaxPool2d(2×2)
	Conv3	Conv(64, 128, 3×3) + ReLU
	Conv4	Conv(128, 128, 3×3) + ReLU
	Pool2	MaxPool2d(2×2)
	Conv5	Conv(128, 256, 3×3) + ReLU
	Conv6	Conv(256, 256, 3×3) + ReLU
	Conv7	Conv(256, 256, 3×3) + ReLU
	Conv8	Conv(256, 256, 3×3) + ReLU
Decoder	Pool3	MaxPool2d(2×2)
	Conv9	Conv(256, 512, 3×3) + ReLU
	AdaSN	Adaptive IN and LN (ρ_s control)
	Conv10	Conv(512, 256, 3×3) + ReLU
	Upsample1	Nearest neighbor upsampling ($\times 2$)
	Conv11	Conv(256, 256, 3×3) + ReLU
	Conv12	Conv(256, 256, 3×3) + ReLU
	Conv13	Conv(256, 256, 3×3) + ReLU
	Conv14	Conv(256, 128, 3×3) + ReLU
	Upsample2	Nearest neighbor upsampling ($\times 2$)
Overall	Conv15	Conv(128, 128, 3×3) + ReLU
	Conv16	Conv(128, 64, 3×3) + ReLU
	Upsample3	Nearest neighbor upsampling ($\times 2$)
	Conv17	Conv(64, 64, 3×3) + ReLU
Overall	Conv18	Conv(64, 3, 3×3) (Output layer)
	-	Encoder $\rightarrow$ AdaSN $\rightarrow$ Decoder

alization performance and domain alignment of our generated data, as reflected by the final test accuracy.

d) *Experimental Scenarios*: To validate the effectiveness of the proposed small-sample generation strategy, we design two benchmark experimental scenarios. In both cases, the CWRU dataset is used as the reference machine to provide fault content information, while different datasets are selected as the monitored machine to evaluate cross-machine generalization. The PCASTNet model is trained to generate synthetic diagnostic samples that match the style of the monitored machine.

- **Scenario I (CWRU to BJTU-RAO)**: The CWRU dataset provides fault content features, while BJTU-RAO contributes style information. PCASTNet generates BJTU-style samples from CWRU semantics.
- **Scenario II (CWRU to HUSTbearing)**: Similarly, PCASTNet learns to synthesize HUST-style samples using the same CWRU fault content.

The classification accuracy on real monitored samples serves as the key indicator of style alignment and diagnostic effectiveness.

C. Scenario I: CWRU to BJTU-RAO

This experimental scenario adopts the CWRU dataset as the reference machine and the BJTU-RAO dataset as the monitored machine. The goal is to evaluate the effectiveness of the proposed sample generation strategy under a cross-machine condition mismatch. All experiments follow the same design, hyperparameter settings, and evaluation metrics as used throughout the section for consistency and comparability.

1) *Hyperparameter Sensitivity Analysis*: To evaluate the model's robustness to its hyperparameter settings, a sensitivity analysis was conducted on the weights of the content, style, and energy losses ($\lambda_c, \lambda_s, \lambda_e$), with each weight being selected from the set $\{1, 5, 10\}$. This range was chosen to represent significant variations in the training objective.

The results are visualized in Fig. 6. Across all 27 combinations tested within this range, the diagnostic accuracy remained consistently high, with the total fluctuation between the maximum and minimum observed accuracy being less than 5%. This minimal performance variation, despite substantial changes in the loss weights, strongly indicates that PCASTNet is robust and not overly sensitive to the precise tuning of these hyperparameters. Such stability is a highly desirable characteristic for practical applications, as it reduces the need for extensive, fine-grained parameter optimization.

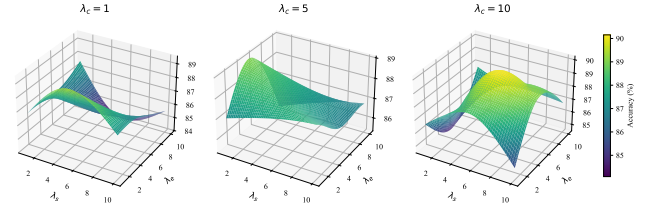


Fig. 6. Interpolated diagnostic accuracy surfaces under different λ_c settings. The plots visualize the model's stable performance, showing that accuracy fluctuates by less than 5% across the tested hyperparameter range.

2) *Comparison with GAN-based Sample Generation Methods*: To evaluate the effectiveness of PCASTNet in generating diagnostic samples, we compare it with several representative GAN-based style transfer methods, including DualGAN, CycleGAN, DiscoGAN, and NiceGAN. All models are trained under consistent configurations, using 50 style samples from the monitored machine and 500 content samples from the reference machine. For fairness, the key hyperparameters (e.g., learning rate, batch size, optimizer settings) of the GAN-based baselines follow the recommended values in their original papers. We performed limited validation to ensure stable convergence on our dataset, but avoided extensive task-specific tuning, thereby balancing reproducibility and fairness in comparison.

DualGAN [27] introduces a dual learning framework with two generators and two discriminators to perform bidirectional translation between unpaired domains using adversarial and reconstruction losses.

CycleGAN [28] leverages a cycle-consistency loss to ensure that a forward-backward translation reconstructs the original input, thereby enabling stable mapping between unpaired domains.

DiscoGAN [29] focuses on discovering cross-domain relations by enforcing both image-level and latent-level consistency, which enhances semantic alignment in style transfer tasks.

NiceGAN [30] improves training efficiency by reusing the discriminator as a feature extractor, reducing model complex-

TABLE II
PERFORMANCE COMPARISON WITH GAN-BASED METHODS (CWRU → BJTU-RAO)

Method	Accuracy (%)	F1 Score	AUC	Convergence Epochs
No augmentation + 50 monitored samples	77.12 ± 2.50	0.744 ± 0.0300	0.942 ± 0.0100	30 ± 10
450 reference samples + 50 monitored samples	86.08 ± 0.40	0.860 ± 0.0000	0.970 ± 0.0000	42 ± 25
DualGAN + 50 monitored samples	85.80 ± 1.00	0.856 ± 0.0150	0.980 ± 0.0000	30 ± 10
CycleGAN + 50 monitored samples	87.24 ± 0.50	0.866 ± 0.0050	0.980 ± 0.0000	42 ± 15
DiscoGAN + 50 monitored samples	86.36 ± 1.30	0.862 ± 0.0150	0.976 ± 0.0050	22 ± 5
NiceGAN + 50 monitored samples	87.12 ± 0.70	0.870 ± 0.0100	0.974 ± 0.0050	20 ± 0
PCASTNet (Ours) + 50 monitored samples	89.48 ± 0.80	0.904 ± 0.0150	0.984 ± 0.0000	24 ± 5

Note: Standard deviation of 0.0000 indicates identical prediction results across multiple runs.

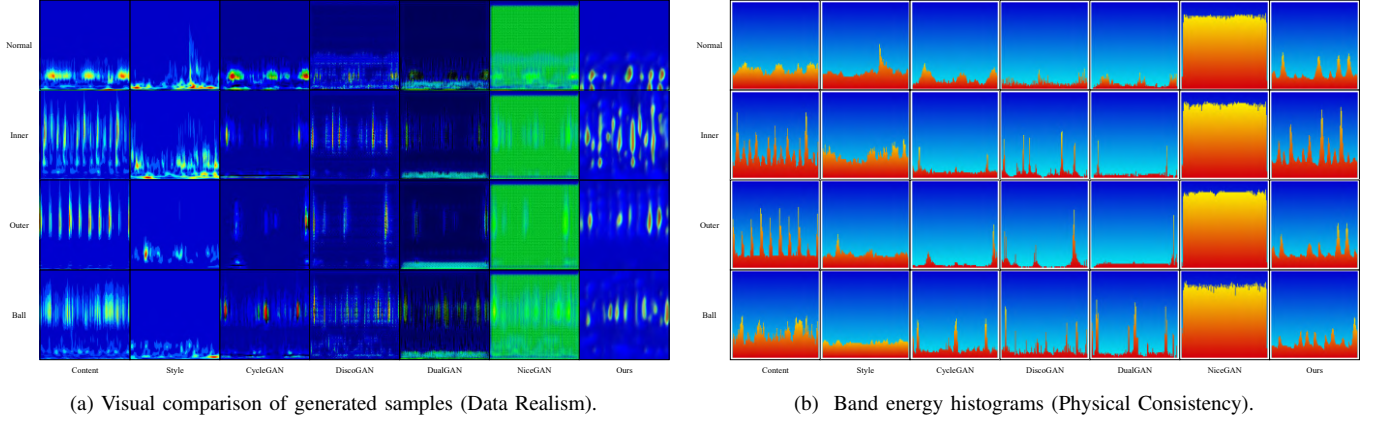


Fig. 7. Qualitative and quantitative comparison of samples generated by different methods. (a) shows the visual quality and realism of the generated spectrograms. (b) shows the corresponding band energy histograms, which reflect the underlying physical energy distribution, allowing for a quantitative assessment of consistency.

ity and enhancing sample quality in unpaired translation.

As shown in Tab. II, PCASTNet consistently outperforms all compared methods in terms of classification accuracy, F1 score, and convergence stability. While certain GAN-based methods such as CycleGAN and DualGAN achieve relatively high AUC scores, they exhibit limitations in semantic preservation and require prolonged training to converge. In contrast, PCASTNet achieves the best overall trade-off between diagnostic performance and computational efficiency, thereby demonstrating its superiority for small-sample, cross-machine fault diagnosis.

While the numerical improvements in Tab. II are modest, the qualitative and physical fidelity of the generated samples, as shown in Fig. 7, reveals the distinct advantages of PCASTNet. Qualitatively, as demonstrated in Fig. 7a, GAN-based methods often introduce artifacts, such as blurred features that obscure fault signatures and unnatural texture noise. In stark contrast, PCASTNet generates samples with superior structural clarity and style consistency, preserving the diagnostic content with high fidelity and thus producing more realistic data. Beyond mere visual realism, PCASTNet also ensures physical consistency, a core contribution of our work. This is quantitatively validated in Fig. 7b through an analysis of band energy histograms, which reflect the underlying energy distribution. The histograms for GAN-generated samples are often erratic and fail to match the target machine's energy patterns, whereas PCASTNet produces histograms that are sig-

nificantly more stable and consistent. Therefore, the true value of PCASTNet lies not just in its superior accuracy but in its generation of diagnostically physically interpretable samples. This combination of data realism, physical consistency, and stable performance underscores its novelty and practical value for industrial applications.

3) *Ablation on Encoder Training Strategies:* We compare two training strategies for the encoder: training on the complete dataset versus training on only 50 selected samples.

Tab. III show that focusing training on the smaller, more homogeneous dataset improves the encoder's ability to capture relevant diagnostic features.

4) *Ablation on Loss Function Combinations:* To evaluate the diagnostic effectiveness of the proposed band energy loss $\mathcal{L}_{\text{energy}}$, we conducted a series of loss-combination ablation experiments under the CWRU to BJTU-RAO scenario. The results are summarized in Tab. IV.

In addition to $\mathcal{L}_{\text{energy}}$, we consider two popular losses from image generation literature as baselines: perceptual loss $\mathcal{L}_{\text{percep}}$ [45] and total variation loss $\mathcal{L}_{\text{tv}}$ [46]. The perceptual loss is formulated as:

$$\mathcal{L}_{\text{percep}}(\mathbf{p}, \mathbf{x}) = \frac{1}{C_l H_l W_l} \|\phi_l(\mathbf{x}) - \phi_l(\mathbf{p})\|_2^2, \quad (15)$$

where $\mathbf{x}$ is the output image, $\mathbf{p}$ is the target image, $\phi_l(\cdot)$ denotes the l -th layer activation from a pretrained VGG network, and C_l , H_l , W_l represent the channel, height, and

TABLE III
ABLATION STUDY: ENCODER TRAINING STRATEGIES (CWRU $\rightarrow$ BJTU-RAO)

Encoder Strategy	Accuracy (%)	F1 Score	AUC	Convergence Epochs
500 reference + 50 monitored samples	87.44 $\pm$ 1.60	0.870 $\pm$ 0.0150	0.982 $\pm$ 0.0050	22 $\pm$ 5
Only 50 monitored samples	88.48 $\pm$ 0.80	0.884 $\pm$ 0.0150	0.980 $\pm$ 0.0000	24 $\pm$ 5

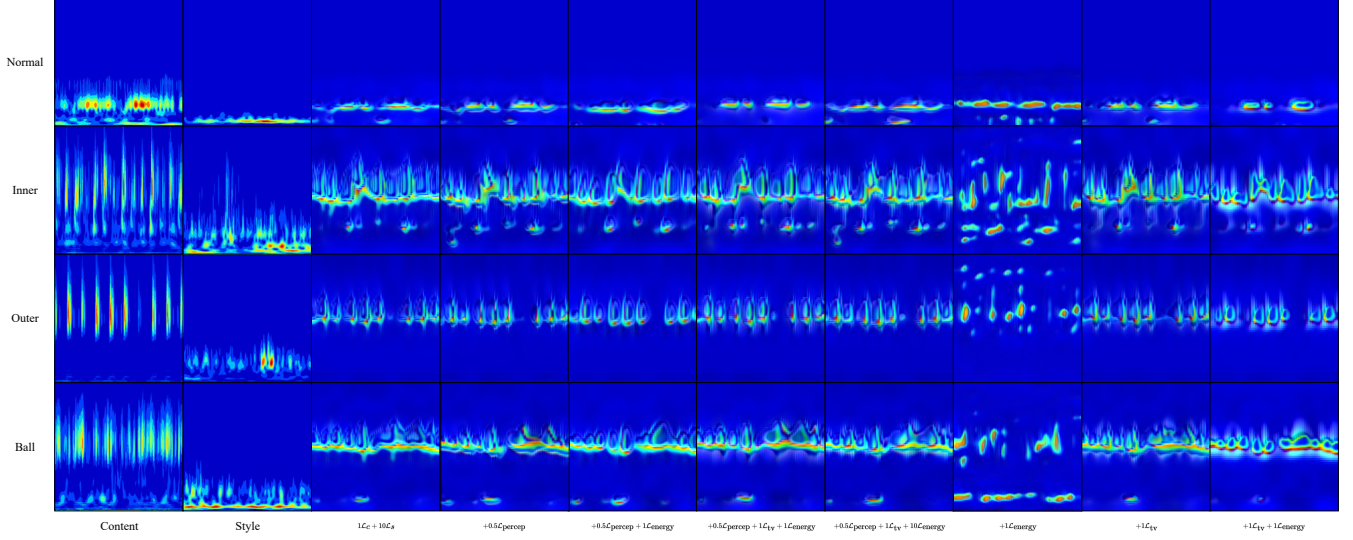


Fig. 8. Visual comparison of generated spectrograms under different loss function combinations. The first two columns show the input Content (CWRU) and target Style (BJTU-RAO) images. Subsequent columns show the generated results with different loss terms added. Note how the addition of $\mathcal{L}_{\text{energy}}$ (e.g., in column '+1 $\mathcal{L}_{\text{energy}}$ ') is crucial for aligning the horizontal high-energy frequency bands, ensuring physical plausibility, a result not achieved by other losses like $\mathcal{L}_{\text{percep}}$ or $\mathcal{L}_{\text{tv}}$, which can introduce blurring or distortion.

width of the feature map respectively. The total variation loss penalizes abrupt pixel changes and is defined as:

$$\mathcal{L}_{\text{tv}}(\mathbf{x}) = \sum_{i,j} \left((x_{i+1,j} - x_{i,j})^2 + (x_{i,j+1} - x_{i,j})^2 \right). \quad (16)$$

In the baseline setting ($1\mathcal{L}_c + 10\mathcal{L}_s$), the test accuracy reached 88.36%.

To provide a more intuitive understanding, we analyze the visual results in Fig. 8 alongside the quantitative data. The perceptual loss ($\mathcal{L}_{\text{percep}}$) and total variation loss ($\mathcal{L}_{\text{tv}}$) are ill-suited for this task. As seen in the figure, $\mathcal{L}_{\text{percep}}$ can introduce visual artifacts and distort the clear impulse patterns of the fault signature. Similarly, $\mathcal{L}_{\text{tv}}$ excessively smooths the spectrogram, blurring the sharp, high-energy frequency lines that are critical for diagnosis. These visual degradations explain their negative impact on diagnostic accuracy.

In stark contrast, our proposed band energy loss ($\mathcal{L}_{\text{energy}}$) is tailored to the problem. Visually, its effect is striking: it forces the generated spectrogram to align its primary energy distribution (the bright horizontal bands) with that of the target style image. This physically meaningful constraint ensures the generated fault signatures are not just texturally similar but are also spectrally consistent with the target machine. This direct alignment of characteristic frequencies explains why configurations including $\mathcal{L}_{\text{energy}}$ (most notably, the '+1 $\mathcal{L}_{\text{energy}}$ ' column, which achieved the highest accuracy of 89.46%) produce both the most realistic-looking samples and the best diagnostic performance. These results demonstrate that the

band energy loss is both effective and modular, making it a critical component of the proposed framework.

These results demonstrate that the band energy loss is both effective and modular, making it a critical component of the proposed framework.

As illustrated in Fig. 9, the incorporation of the band energy loss ($\mathcal{L}_{\text{energy}}$) not only improves the overall classification performance but also leads to a particularly significant boost for the B fault type. This is evidenced by a sharper increase in the diagonal element of the confusion matrix for the B fault, indicating enhanced discriminability and reduced misclassification compared to other loss combinations.

5) *Comparison of Normalization Modules*: The core of our proposed method lies in the AdaSN module, which adaptively blends style statistics using a learnable parameter ρ_s . To systematically validate our design choices and understand the roles of content- and style-based adaptation, we designed three alternative normalization modules for an in-depth ablation study: **AdaIN**, **Adaptive Content Normalization (AdaCN)**, and **Adaptive Content and Style Normalization (AdaCSN)**.

- **AdaIN** [41] serves as our baseline. Its formulation, which uses fixed instance-level statistics from the style features, is defined in Eq. (3).
- **AdaCN** is a variant we designed to explore content-adaptive normalization. It modifies the normalization part of AdaIN by computing adaptive statistics, $\gamma_c(F_c)$ and $\beta_c(F_c)$, from the content features themselves using a

TABLE IV
ABLATION ON LOSS FUNCTION COMBINATIONS (CWRU → BJTU-RAO)

Loss Combination	Accuracy (%)	F1 Score	AUC	Convergence Epochs
$1\mathcal{L}_c + 10\mathcal{L}_s$	88.36 ± 1.10	0.880 ± 0.015	0.980 ± 0.000	32 ± 10
$+0.5\mathcal{L}_{\text{percep}}$	87.80 ± 1.40	0.874 ± 0.015	0.980 ± 0.000	20 ± 0
$+0.5\mathcal{L}_{\text{percep}} + 1\mathcal{L}_{\text{energy}}$	87.72 ± 2.00	0.874 ± 0.020	0.978 ± 0.005	26 ± 10
$+0.5\mathcal{L}_{\text{percep}} + 1\mathcal{L}_{\text{tv}} + 1\mathcal{L}_{\text{energy}}$	87.44 ± 3.50	0.872 ± 0.040	0.978 ± 0.005	24 ± 5
$+0.5\mathcal{L}_{\text{percep}} + 1\mathcal{L}_{\text{tv}} + 10\mathcal{L}_{\text{energy}}$	89.00 ± 1.10	0.890 ± 0.010	0.980 ± 0.010	30 ± 15
$+1\mathcal{L}_{\text{energy}}$	89.46 ± 1.60	0.901 ± 0.015	0.986 ± 0.005	20 ± 0
$+1\mathcal{L}_{\text{tv}}$	88.76 ± 1.20	0.902 ± 0.055	0.980 ± 0.000	20 ± 0
$+1\mathcal{L}_{\text{tv}} + 1\mathcal{L}_{\text{energy}}$	87.16 ± 0.60	0.870 ± 0.010	0.978 ± 0.005	30 ± 15

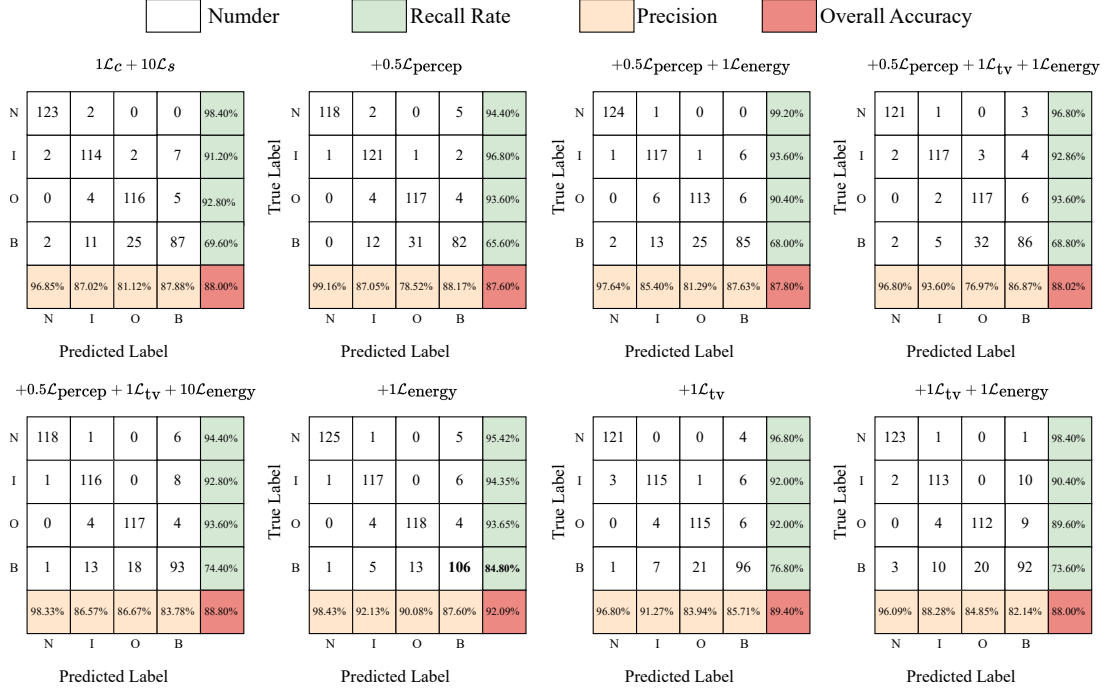


Fig. 9. Detailed confusion matrices under different loss function combinations in the CWRU→BJTU-RAO task. Each cell displays the raw count of predictions, while the values on the right and bottom axes represent per-class precision and recall, respectively. The overall accuracy is shown in the bottom-right corner of each matrix. Each sub-figure corresponds to an ablation setting from Tab. IV: (a) $1\mathcal{L}_c + 10\mathcal{L}_s$, (b) $+0.5\mathcal{L}_{\text{percep}}$, (c) $+0.5\mathcal{L}_{\text{percep}} + 1\mathcal{L}_{\text{energy}}$, (d) $+0.5\mathcal{L}_{\text{percep}} + 1\mathcal{L}_{\text{tv}} + 1\mathcal{L}_{\text{energy}}$, (e) $+0.5\mathcal{L}_{\text{percep}} + 1\mathcal{L}_{\text{tv}} + 10\mathcal{L}_{\text{energy}}$, (f) $+1\mathcal{L}_{\text{energy}}$, (g) $+1\mathcal{L}_{\text{tv}}$, (h) $+1\mathcal{L}_{\text{tv}} + 1\mathcal{L}_{\text{energy}}$. In (f), the number of correctly predicted Ball fault samples and its accuracy are highlighted in bold to emphasize the effect of the energy loss.

learnable parameter ρ_c :

$$\gamma_c(F_c) = \rho_c \sigma_{\text{IN}}(F_c) + (1 - \rho_c) \sigma_{\text{LN}}(F_c). \quad (17)$$

$$\beta_c(F_c) = \rho_c \mu_{\text{IN}}(F_c) + (1 - \rho_c) \mu_{\text{LN}}(F_c). \quad (18)$$

The full transformation is then:

$$\text{AdaCN}(F_c, F_s) = \sigma_{\text{IN}}(F_s) \cdot \left(\frac{F_c - \beta_c(F_c)}{\gamma_c(F_c)} \right) + \mu_{\text{IN}}(F_s). \quad (19)$$

- **AdaSN (Ours)** is our proposed module, which focuses on style-adaptive affine transformation. As defined in Eq. (4), it generates adaptive affine parameters $\gamma_s(F_s)$ and $\beta_s(F_s)$ by blending IN and LN statistics from the style features, with the blending mechanism detailed in Eq. (5) and (6).
- **AdaCSN** is the final variant, combining both mechanisms. It employs the content-adaptive normalization

from AdaCN and the style-adaptive affine transformation from AdaSN:

$$\text{AdaCSN}(F_c, F_s) = \gamma_s(F_s) \cdot \left(\frac{F_c - \beta_c(F_c)}{\gamma_c(F_c)} \right) + \beta_s(F_s). \quad (20)$$

We evaluate these four normalization strategies—AdaIN, AdaCN, AdaSN (Ours), and AdaCSN—on the CWRU to BJTU-RAO transfer task.

As shown in Tab. V, AdaSN achieves the highest accuracy (88.48%) and F1 score (0.884), with stable convergence (epoch std = 5), indicating superior performance. AdaIN, though parameter-free, provides competitive accuracy due to straightforward statistical alignment, while AdaCN demonstrates greater variance, likely due to sensitivity in content modulation. AdaCSN offers a balance between accuracy and stability. Fig. 10 further visualizes training loss dynamics,

TABLE V
PERFORMANCE COMPARISON OF NORMALIZATION MODULES (CWRU $\rightarrow$ BJTU-RAO)

The Normalized Module Used	Accuracy (%)	F1 Score	AUC	Convergence Epochs
AdaIN	87.36 ± 1.20	0.870 ± 0.0150	0.976 ± 0.0050	20 ± 0
AdaCN	85.88 ± 1.20	0.854 ± 0.0150	0.976 ± 0.0050	24 ± 10
AdaSN (Ours)	88.48 ± 1.20	0.884 ± 0.0150	0.980 ± 0.0000	24 ± 5
AdaCSN	87.00 ± 0.50	0.866 ± 0.0050	0.980 ± 0.0000	22 ± 5

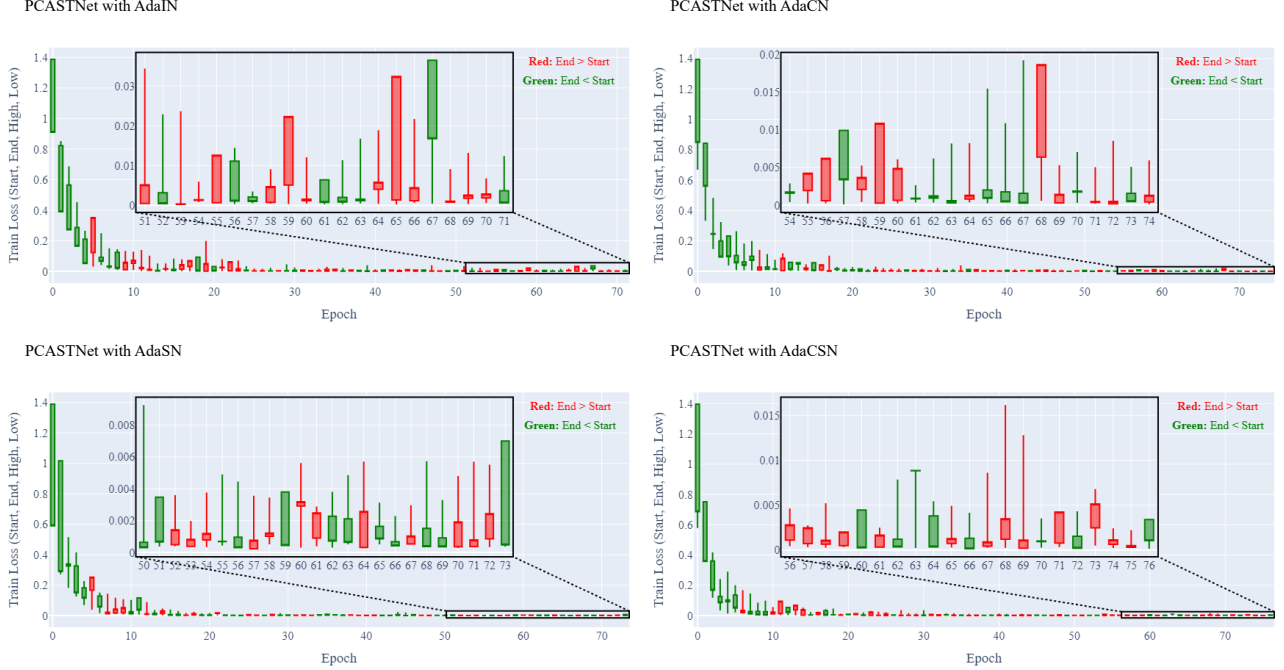


Fig. 10. Training loss candlestick curves for four normalization modules on the CWRU to BJTU-RAO task. For each module, the candlestick plot visualizes the range and dynamics of training loss at each epoch. The **red** bar indicates epochs where the loss increased (End > Start), while the **green** bar marks epochs where the loss decreased (End < Start). Each candlestick encodes **Start**, **End**, **High**, and **Low** loss values during the epoch, intuitively revealing convergence patterns and oscillations. The right inset in each sub-figure presents a zoom-in of the last 20 epochs, highlighting training stability and convergence behavior among different normalization modules.

where AdaSN exhibits smoother convergence and less oscillation, confirming its training stability.

In conclusion, AdaSN offers an effective balance between style regulation and diagnostic consistency, making it a key normalization mechanism for small-sample, cross-machine fault diagnosis tasks.

D. Scenario II: CWRU to HUSTbearing

To further evaluate the generalization capability of the proposed method, we consider a distinct diagnostic setting where the HUSTbearing dataset—featuring different sensor configurations, bearing types, and operational conditions—is used for testing, while synthetic samples are still generated from CWRU data.

All evaluation protocols and metrics are consistent with those in Scenario I to enable a fair and systematic comparison.

1) *Comparison with GAN-based Sample Generation Methods*: As reported in Tab. VI, PCASTNet again achieves the best overall performance, outperforming all GAN-based

baselines in accuracy (84.24%), F1 score (0.844), and AUC (0.972), and converges quickly (26 ± 5 epochs). While CycleGAN and DiscoGAN deliver relatively stable results, they fall short in capturing fine-grained diagnostic patterns under small-sample constraints. NiceGAN exhibits higher variance in both predictive metrics and convergence, and DualGAN and DiscoGAN are prone to producing blurred features and unstable distributions.

These results corroborate the findings from the Scenario I. The integration of AdaSN and physics-constrained energy alignment enables PCASTNet to generate physically meaningful and semantically aligned fault representations. This advantage is further accentuated under challenging cross-machine conditions, confirming PCASTNet's strong generalization and practical applicability.

The overall quantitative performance is further visualized in Fig. 11, highlighting the superiority of PCASTNet across both scenarios.

2) *Ablation on Encoder Training Strategy*: The influence of training data composition on the encoder is examined in

TABLE VI
PERFORMANCE COMPARISON WITH GAN-BASED METHODS (CWRU $\rightarrow$ HUSTBEARING)

Method	Accuracy (%)	F1 Score	AUC	Convergence Epochs
No augmentation + 50 monitored samples	71.60 $\pm$ 0.57	0.700 $\pm$ 0.0071	0.948 $\pm$ 0.0045	54 $\pm$ 25
450 reference samples + 50 monitored samples	80.56 $\pm$ 0.89	0.806 $\pm$ 0.0114	0.940 $\pm$ 0.0000	32 $\pm$ 5
CycleGAN + 50 monitored samples	81.84 $\pm$ 2.85	0.820 $\pm$ 0.0283	0.964 $\pm$ 0.0089	38 $\pm$ 20
NiceGAN + 50 monitored samples	80.48 $\pm$ 4.37	0.804 $\pm$ 0.0434	0.956 $\pm$ 0.0261	36 $\pm$ 15
DiscoGAN + 50 monitored samples	81.88 $\pm$ 0.69	0.820 $\pm$ 0.0071	0.964 $\pm$ 0.0055	32 $\pm$ 15
DualGAN + 50 monitored samples	81.12 $\pm$ 1.73	0.812 $\pm$ 0.0192	0.948 $\pm$ 0.0045	34 $\pm$ 10
PCASTNet (Ours) + 50 monitored samples	84.24 $\pm$ 0.54	0.844 $\pm$ 0.0089	0.972 $\pm$ 0.0045	26 $\pm$ 5

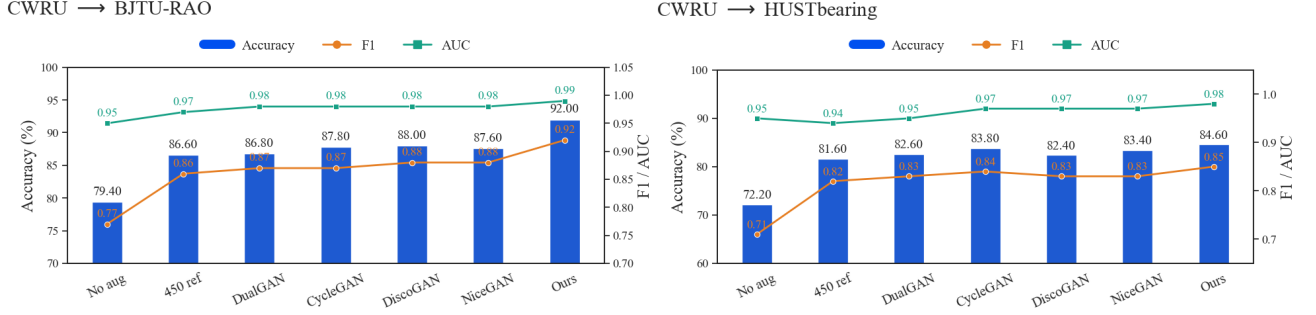


Fig. 11. Comparison of all methods: Accuracy (bar), F1 score and AUC (line). Left: CWRU $\rightarrow$ BJTU-RAO; Right: CWRU $\rightarrow$ HUSTbearing.

TABLE VII
ABLATION STUDY: ENCODER TRAINING STRATEGIES (CWRU $\rightarrow$ HUSTBEARING)

Encoder Strategy	Accuracy (%)	F1 Score	AUC	Convergence Epochs
500 reference + 50 monitored samples	81.12 $\pm$ 3.10	0.812 $\pm$ 0.030	0.942 $\pm$ 0.010	32 $\pm$ 10
Only 50 monitored samples	84.24 $\pm$ 0.60	0.844 $\pm$ 0.010	0.972 $\pm$ 0.005	26 $\pm$ 5

TABLE VIII
ABLATION ON LOSS FUNCTION COMBINATIONS (CWRU $\rightarrow$ HUSTBEARING)

Loss Combination	Accuracy (%)	F1 Score	AUC	Convergence Epochs
$1\mathcal{L}_c + 10\mathcal{L}_s$	83.58 $\pm$ 1.35	0.834 $\pm$ 0.0150	0.970 $\pm$ 0.0000	24 $\pm$ 5
$+0.5\mathcal{L}_{\text{percep}}$	82.56 $\pm$ 0.80	0.826 $\pm$ 0.0100	0.962 $\pm$ 0.0050	20 $\pm$ 0
$+0.5\mathcal{L}_{\text{percep}} + 1\mathcal{L}_{\text{energy}}$	83.36 $\pm$ 0.85	0.834 $\pm$ 0.005	0.9700 $\pm$ 0.0000	36 $\pm$ 5
$+0.5\mathcal{L}_{\text{percep}} + 1\mathcal{L}_{\text{tv}} + 1\mathcal{L}_{\text{energy}}$	82.68 $\pm$ 0.90	0.826 $\pm$ 0.0050	0.966 $\pm$ 0.0050	26 $\pm$ 10
$+0.5\mathcal{L}_{\text{percep}} + 1\mathcal{L}_{\text{tv}} + 10\mathcal{L}_{\text{energy}}$	82.44 $\pm$ 2.00	0.822 $\pm$ 0.0200	0.970 $\pm$ 0.0100	36 $\pm$ 15
$+1\mathcal{L}_{\text{energy}}$	84.24 $\pm$ 0.60	0.844 $\pm$ 0.0100	0.972 $\pm$ 0.0050	26 $\pm$ 5
$+1\mathcal{L}_{\text{tv}}$	82.52 $\pm$ 0.10	0.820 $\pm$ 0.0000	0.970 $\pm$ 0.0000	34 $\pm$ 10
$+1\mathcal{L}_{\text{tv}} + 1\mathcal{L}_{\text{energy}}$	82.16 $\pm$ 0.40	0.818 $\pm$ 0.0050	0.970 $\pm$ 0.0000	22 $\pm$ 5

Tab. VII. Interestingly, training solely on the small monitored set outperforms the full-data strategy, possibly due to better feature alignment under limited-domain variation.

3) *Ablation on Loss Function Combinations*: Tab VIII suggests that enforcing high-level semantic similarity and spatial continuity may interfere with the preservation of physically meaningful spectral energy in the HUSTbearing dataset, where signal acquisition differs significantly from the reference platform. Like the results in Scenario I, the combination of perceptual and total variation losses fails to produce the highest accuracy in this setting. In contrast, configurations involving only the energy constraint exhibit more stable results, reaffirming its pivotal role in maintaining physical integrity during style transfer.

4) *Comparison of Normalization Modules*: Results in Tab. IX show all four normalization modules remain competitive. AdaSN again leads with the best accuracy (84.24%) and F1 score (0.844), maintaining strong convergence stability. AdaCSN achieves comparable accuracy but with slightly higher epoch variance, reflecting increased instability under complex domain shifts. AdaIN and AdaCN yield marginally lower scores, with AdaCN in particular showing higher F1 standard deviation, indicating greater sensitivity to style modulation.

These findings are consistent with those from the CWRU to BJTU-RAO setting, reinforcing AdaSN's effectiveness for balancing generalization and semantic preservation, especially in low-resource or highly heterogeneous environments. The

TABLE IX
PERFORMANCE COMPARISON OF NORMALIZATION MODULES (CWRU $\rightarrow$ HUSTBEARING)

The Normalized Module Used	Accuracy (%)	F1 Score	AUC	Convergence Epochs
AdaIN	83.92 $\pm$ 0.18	0.840 $\pm$ 0.0000	0.970 $\pm$ 0.0000	26 $\pm$ 5
AdaCN	82.60 $\pm$ 1.30	0.826 $\pm$ 0.0167	0.966 $\pm$ 0.0055	20 $\pm$ 0
AdaSN (Ours)	84.24 $\pm$ 0.54	0.844 $\pm$ 0.0089	0.972 $\pm$ 0.0045	26 $\pm$ 5
AdaCSN	84.20 $\pm$ 0.45	0.840 $\pm$ 0.0071	0.970 $\pm$ 0.0000	30 $\pm$ 10

adaptive style control provided by ρ_s allows AdaSN to deliver scalable performance for cross-machine fault diagnosis.

E. Summary

As demonstrated in the above two scenarios, PCASTNet consistently outperforms other sample generation methods, achieving the highest diagnostic accuracy with an average improvement of 2.6%–5.8%. In addition, it exhibits faster convergence and lower performance variance, indicating enhanced training stability and generalization capability. In particular, the proposed AdaSN module consistently outperforms other normalization designs on both BJTU-RAO and HUSTbearing platforms, validating its effective balance between style control and physical fidelity. In the comparison of encoder training strategies, results suggest that training solely on small-sample sets improves feature alignment when the test distribution is relatively stable, thereby promoting more stable and effective learning. Furthermore, the proposed band energy loss $\mathcal{L}_{\text{energy}}$ proves essential across all tasks. It reinforces the preservation of diagnostic frequency components and achieves physical consistency without relying on explicit prior models.

Overall, PCASTNet integrates multi-level structural designs and physics-guided constraints to deliver high-performance fault sample generation under small-sample, cross-machine conditions, offering a stable and effective paradigm for intelligent fault diagnosis in practical applications.

IV. CONCLUSION

To tackle the problem of diagnostic performance degradation caused by structural variation and insufficient fault samples in cross-machine settings, this paper proposes a Physics-Constrained Adaptive Style Transfer Network (PCASTNet) to synthesize realistic vibration samples for the monitored machine. The method utilizes wavelet-based time–frequency representations and a learnable normalization module to decouple diagnostic semantics and machine-specific styles, while enforcing spectral consistency through a band energy constraint. Experimental results show that PCASTNet significantly improves diagnostic accuracy under small-sample conditions. The proposed strategy offers a generalizable and physically interpretable solution for intelligent fault diagnosis in practical mechanical systems.

The limitation lies in the static configuration of wavelet parameters, which may reduce adaptability to signals with strong nonstationarity. Future work may explore dynamic wavelet selection, lightweight self-supervised adaptation and one-dimensional CNN-based style transfer network to enhance deployment flexibility and reduce manual tuning effort.

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